

Binary Reverse Engineering Assessment

String Deobfuscation Challenge

Question: Reverse Engineer Password from Obfuscated Binary

Your Task

You are provided with a compiled Windows executable "**Crackme.exe**" that prompts for a password. The password is hidden using simple obfuscation techniques. Use reverse engineering tools to discover the correct password and demonstrate successful authentication.

Binary Behavior

The "**Crackme.exe**" program will:

- Prompt: "Enter password: "
- Accept user input
- Display "Wrong password!" for incorrect input
- Display "Correct password. Welcome!" for correct input

Required Analysis

Reverse Engineering Process:

1. Load the "**Crackme.exe**" binary in a decompiler (IDA Pro, Ghidra, or x64dbg)
2. Analyze the main function and password checking logic
3. Identify the obfuscated string data
4. Understand the deobfuscation algorithm
5. Calculate the original password
6. Verify by running the binary with the discovered password

Required Deliverables

1. Decompiler Analysis Screenshots

- Main function disassembly/decompiled code
- Obfuscated string array in memory/data section
- Deobfuscation logic (the algorithm used to reveal the password)
- Password comparison function

2. Manual Password Recovery

- Step-by-step calculation showing how you decoded the password
- Explanation of the obfuscation method used
- Final decoded password string

3. Execution Verification

- Screenshot of running the binary with incorrect password
- Screenshot of running the binary with correct password showing success message

Expected Skills Demonstrated

Static Analysis:

- Ability to navigate decompiler interface
- Understanding of assembly/decompiled C code
- Recognition of string obfuscation patterns
- Mathematical calculation of decoded values

Dynamic Verification:

- Successful execution of the target binary
- Proof of correct password discovery

Submission Requirements

- **Files:** Decompiler screenshots, calculation worksheet
- **Evidence:** Binary execution screenshots showing success

 **BINARY EXECUTION REQUIREMENT:** You must successfully run the “**Crackme.exe**” binary and demonstrate the correct password works by showing the success message.

Prerequisites: Basic reverse engineering skills, decompiler usage (IDA/Ghidra/x64dbg), understanding of string manipulation and basic encoding